

Shaun John Thomas

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PROFILE

Final-year Aero-Mechanical Engineer with strong fundamentals in structural design and aerodynamics, and hands-on experience in CAD, FEA and manufacturing. Currently applying this to the design, upcoming fabrication and flight testing of a Blended Wing Body Unmanned Aerial Vehicle (UAV). Directed a 4-member team through the design, Geometric Dimensioning and Tolerancing (GD&T)-based manufacturing and build of a 10,000 ft model rocket, coordinating sub-systems from structures to propulsion. Proficient across the full simulation-to-hardware pipeline, with proven cross-functional collaboration in multidisciplinary student engineering teams.

EXPERIENCE AND PROJECTS

Airframe Design Engineer

Glasgow, UK | September 2025 - Present

StrathAIS, Student Rocketry Team, University of Strathclyde

- Designed bulkhead and supporting structures in SolidWorks, including airframe attachment bolt pattern for a Mach 1+ rocket, evidenced through PDR and CDR reviews. The team placed 8th/21 in the COTS 30K at IREC 2026 with a 24,528 ft apogee.
- Ran FEA simulations to shape critical design choices, reducing bulkhead weight by over 50% while preserving structural margins.
- Recommended and implemented changes in manufacturing decisions to minimize error and material waste with the experience gained from previous failures.

Blended Wing Body UAV Project

Glasgow, UK | August 2026 - Present

Individual Project

- Evaluated initial BWB geometries using a Clark Y airfoil in XFLR5, identified a destabilizing -0.105 pitching moment, and pivoted to an MH60 profile achieving near-neutral trim ($C_m = 0.01$).
- Redesigned geometric parameters in XFLR5, balancing payload volume against lift/drag targets. The MH-series profile achieved higher overall efficiency than Clark Y despite requiring a higher trim AoA (5° vs. 2°).
- Validating the optimized MH-series configuration through CFD sweeps in ANSYS, targeting a 5–8% MAC static margin and reduced induced drag prior to fabrication.
- Developing a physical model using 3D-printed structures, CNC tooling, and composite reinforcements. Targeting a sub-2kg structural mass ahead of flight testing in February.

Airframe Lead

Chennai, India | August 2023 – June 2025

Team Ignition, Student Rocketry Team, Vellore Institute of Technology

- Directed a team of 4 to design and build the airframe of a model rocket to reach an altitude of 10,000 feet while collaborating closely with various departments to reach innovative design compromises across the project.
- Validated design choices by running CFD, FEA, pressure and vibration analyses with the help of technical guidance from external manufacturers and mentors.
- Diagnosed the launch failure to a loose avionics connector that melted under heat. The team revised connector mounting and thermal-isolation practices for future rockets.

EDUCATION

University

BEng Aero-Mechanical Engineering, University of Strathclyde
Current Average: 69%

Glasgow, UK | September 2025- Present

BTech Mechanical Engineering, Vellore Institute of Technology (Transferred)
2 Year CGPA: 8.4/10

Chennai, India | August 2023 – June 2025

SKILLS & CERTIFICATIONS

Skills and Soft Skills:

- CAD: SolidWorks, OnShape
- Manufacturing: 3D-Printing, Composites, GD&T
- Technical leadership & cross-department collaboration
- Project Management & Problem Solving
- Simulations: ANSYS (CFD & FEA), XFLR5
- Machining: CNC machining, power tools
- Numerical Analysis: MATLAB & SIMULINK, Python
- Design Reviews & Technical Communication

Certifications: Certified SolidWorks Design Associate (CSWA)